Class 10 · Mathematics

Chapter 4: Special Products and Factorisation

Worksheet — Special Products and Factorisation (Class 10 NIOS Mathematics)

Total marks: **30** · Questions: **11**

Instructions: Attempt all 10 questions in your notebook. Show every step of your working for the subjective questions; choose the single best option for MCQs. Verify each factorisation by re-multiplying. Total time: 50 minutes.

Q1. Expand $(x + 5)^2$. [1 mark]

(A) $x^2 + 25$

(B) $x^2 + 10x + 25$

(C) $x^2 + 5x + 25$

(D) $x^2 + 10x + 10$

Q2. $(a + b)^2 = a^2 + b^2$. [1 mark]

(A) True

(B) False

Q3. $a^2 – b^2 = (a + b)$(____). [1 mark]

Q4. Use $(a + b)(a – b) = a^2 – b^2$ to compute 103×97. [2 marks]

(A) 10000

(B) 9991

(C) 9999

(D) 10009

Q5. Factor $6x^2 + 9x$. [1 mark]

(A) $3(2x^2 + 3x)$

(B) $3x(2x + 3)$

(C) $x(6x + 9)$

(D) $6x(x + 9)$

Q6. Expand $(3y – 2)^2$ step by step. [3 marks]

Q7. Factor by regrouping: $ax + ay + bx + by$. [3 marks]

Q8. Factor $x^2 + 5x + 6$ by finding two numbers with the right sum and product. Verify by re-multiplying. [3 marks]

Q9. Factor $3x^2 + 11x + 6$ by splitting the middle term. Show every step. [4 marks]

Q10. Factor $x^4 – 1$ completely. Identify each pattern you used. [6 marks]

Q11. Factor $12x^2 + 11x – 5$ by splitting the middle term. Show every step and verify by re-multiplying. [5 marks]

Answer Key

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Q1. Expand $(x + 5)^2$.

Answer: $x^2 + 10x + 25$

$(a + b)^2$ with $a = x$, $b = 5$: $x^2 + 2\cdot x\cdot 5 + 5^2 = x^2 + 10x + 25$.

MCQ · EASY · 10-NIOS-MATH-SPF-01

Q2. $(a + b)^2 = a^2 + b^2$.

Answer: False

$(a + b)^2 = a^2 + 2ab + b^2$; the middle term $2ab$ is not zero in general.

TRUE_FALSE · EASY · 10-NIOS-MATH-SPF-01

Q3. $a^2 – b^2 = (a + b)(\_\_\_\_\_)$.

Answer: $a – b$

Difference-of-squares identity.

FILL_BLANK · EASY · 10-NIOS-MATH-SPF-02

Q4. Use $(a + b)(a – b) = a^2 – b^2$ to compute 103×97.

Answer: 9991

$103 \times 97 = (100 + 3)(100 – 3) = 100^2 – 3^2 = 10000 – 9 = 9991$.

MCQ · MEDIUM · 10-NIOS-MATH-SPF-02

Q5. Factor $6x^2 + 9x$.

Answer: $3x(2x + 3)$

HCF of $6x^2$ and $9x$ is $3x$.

MCQ · EASY · 10-NIOS-MATH-SPF-05

Q6. Expand $(3y – 2)^2$ step by step.

Answer: $(a – b)^2$ with $a = 3y$, $b = 2$: $(3y)^2 – 2\cdot(3y)\cdot 2 + 2^2 = 9y^2 – 12y + 4$.

Square-of-difference identity.

SHORT_ANSWER · EASY · 10-NIOS-MATH-SPF-01

Q7. Factor by regrouping: $ax + ay + bx + by$.

Answer: Group: $a(x + y) + b(x + y) = (a + b)(x + y)$.

Each pair pulls out a common factor; the resulting parts share the binomial $(x + y)$.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-SPF-06

Q8. Factor $x^2 + 5x + 6$ by finding two numbers with the right sum and product. Verify by re-multiplying.

Answer: Need $p + q = 5$, $p \cdot q = 6 \rightarrow (2, 3)$. So $x^2 + 5x + 6 = (x + 2)(x + 3)$. Verify: $(x + 2)(x + 3) = x^2 + 3x + 2x + 6 = x^2 + 5x + 6$ ✓.

Type-A quadratic factorisation.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-SPF-07

Q9. Factor $3x^2 + 11x + 6$ by splitting the middle term. Show every step.

Answer: $a \cdot c = 3 \cdot 6 = 18$; need sum 11, product 18 $\rightarrow (2, 9)$. Split $11x = 2x + 9x$: $3x^2 + 2x + 9x + 6 = x(3x + 2) + 3(3x + 2) = (3x + 2)(x + 3)$.

Split-the-middle for $ax^2 + bx + c$.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-SPF-07

Q10. Factor $x^4 – 1$ completely. Identify each pattern you used.

Answer: Step 1 (difference of squares): $x^4 - 1 = (x^2)^2 - 1^2 = (x^2 - 1)(x^2 + 1)$. Step 2 (difference of squares again): $x^2 - 1 = (x - 1)(x + 1)$. Note $x^2 + 1$ has no real factor (sum of squares). Final: $x^4 - 1 = (x - 1)(x + 1)(x^2 + 1)$.

Apply difference of squares iteratively.

LONG_ANSWER · HARD · 10-NIOS-MATH-SPF-08

Q11. Factor $12x^2 + 11x – 5$ by splitting the middle term. Show every step and verify by re-multiplying.

Answer: $a \cdot c = 12 \cdot (-5) = -60$; need sum 11, product $-60 \rightarrow (15, -4)$. Split: $12x^2 + 15x - 4x - 5 = 3x(4x + 5) - 1(4x + 5) = (4x + 5)(3x - 1)$. Verify: $(4x + 5)(3x - 1) = 12x^2 - 4x + 15x - 5 = 12x^2 + 11x - 5$ ✓.

Splitting the middle term works for any $ax^2 + bx + c$.

LONG_ANSWER · HARD · 10-NIOS-MATH-SPF-07